NeuroADHD Final System Architecture & Benchmark Specification

Complete Directory Hierarchy (Tree of Folders), Isolated Environments, Multi-Model Pipeline & Results

1. Executive Summary & System Identity

This specification documents the complete final architecture for the **NeuroADHD** research ecosystem and its reproducible multi-model benchmarking workspace **NeuroADHD_ModelBenchmark**. The system performs automated binary classification of **Attention Deficit Hyperactivity Disorder (ADHD=1)** vs. **Typically Developing Controls (Control=0)** using resting-state functional MRI (rs-fMRI) functional connectivity features derived from the **ADHD-200 Consortium**.

- **Original Core Workspace (NeuroADHD/):** Treated as 100% READ-ONLY and preserved without modification.
- **Primary ML Feature Table:** 153 subjects × 19,900 Fisher-z transformed CC200 ROI connectivity features.
- **Partition Strategy:** Whole-site group holdout (114 Train, 19 NeuroIMAGE Validation, 20 Peking_2 Test).
- **Model Benchmark Workspace (NeuroADHD_ModelBenchmark/):** Isolated multi-model benchmarking environment.
- **Isolated Virtual Environments:** 7 independent Python environments (logistic_regression, linear_svm, rbf_svm, random_forest, extra_trees, xgboost, ensemble).
- **Evaluation & Selection:** Selected top model (Random Forest, Val ROC-AUC = 0.7333) and evaluated ONCE on locked Peking_2 Test set (Test Acc = 0.5500, Spec = 0.8000).

2. Final Workspace Tree of Folders (TOF)

The complete workspace root (/Users/apple/Documents/Codex/2026-09-29/down) contains two primary project directories: the core research project (NeuroADHD/) and the reproducible multi-model benchmark suite (NeuroADHD_ModelBenchmark/):

```
down/
■■■ work_download_adhd200.py [Multithreaded concurrent S3 downloader script]
■■■ generate_pdf_report.py [PDF documentation architecture generator script]
■■■ NeuroADHD_Workspace_Architecture...pdf [Generated publication-quality PDF report]
■■■ work/ [Empty scratch directory for transient execution tasks]
■■■ outputs/ [Raw BIDS dataset destination & S3 inventory]
■ ■■■ ADHD200_RawDataBIDS/ [BIDS formatted NIfTI anatomical and functional volumes]
■ ■ ■■■ Brown/ [Brown University raw fMRI BIDS files]
■ ■ ■■■ KKI/ [Kennedy Krieger Institute raw fMRI BIDS files]
■ ■■■ ADHD200_RawDataBIDS_manifest.tsv [S3 object key & size inventory (2,426 rows)]
■ ■■■ ADHD200_RawDataBIDS_file_list.txt [Plain text listing of S3 relative keys]
■
■■■ NeuroADHD/ [Core Research & Data Lake Directory (READ-ONLY)]
■ ■■■ .venv/ [Local Python 3.11 Virtual Environment]
■ ■■■ configs/splits/ [cc200_site_holdout.json group holdout configuration]
■ ■■■ data/adhd200/ [Multi-stage Data Lake]
■ ■ ■■■ raw/ [Raw ROI signals (cc200, aal, falff, motion, phenotypes)]
■ ■ ■■■ interim/ [Validation reports, schemas, subject label maps]
■ ■ ■■■ features/cc200/ [cc200_connectivity_features.csv (153 × 19,903 matrix)]
■ ■ ■■■ processed/splits/ [subject_split_manifest.csv, train/val/test CSVs]
■ ■■■ src/ [Production Python Modules]
■ ■ ■■■ features/ [extract_cc200_connectivity.py (Fisher-z transform)]
■ ■ ■■■ preprocessing/ [cc200_ml_preprocessing.py, split validators]
■ ■■■ tests/ [test_cc200_preprocessing_leakage.py unit test suite]
■ ■■■ README.md & requirements.txt [Documentation & dependency declarations]
■ ■■■ models/, notebooks/, results/ [Reserved project output directories]
■
■■■ NeuroADHD_ModelBenchmark/ [Isolated Multi-Model Benchmarking Suite]
■■■ configs/ [Master & Model Configuration JSONs]
■ ■■■ benchmark_config.json [Master benchmark configuration file]
■ ■■■ models/ [logistic_regression, linear_svm, rbf_svm, rf, extra_trees, xgboost]
■■■ model_envs/ [7 Model-Specific Isolated Python Virtual Environments]
■ ■■■ logistic_regression/ [Isolated env + requirements + lockfile]
■ ■■■ linear_svm/ [Isolated env + requirements + lockfile]
■ ■■■ rbf_svm/ [Isolated env + requirements + lockfile]
■ ■■■ random_forest/ [Isolated env + requirements + lockfile]
■ ■■■ extra_trees/ [Isolated env + requirements + lockfile]
■ ■■■ xgboost/ [Isolated env + xgboost 3.2.0 + lockfile]
■ ■■■ ensemble/ [Isolated voting & stacking ensemble environment]
■■■ scripts/ [Benchmarking Pipeline Execution Engine]
■ ■■■ preflight_checks.py [Dataset, split, label & zero-leakage validator]
■ ■■■ setup_environments.py [Virtual environment creator & lockfile generator]
■ ■■■ train_model.py [Model training, val grid search & artifact saver]
■ ■■■ evaluate_model.py [Metrics calculator (Acc, F1, ROC-AUC, Spec, Sens, CM)]
■ ■■■ run_all_models.py [Master benchmark orchestrator & test unleasher]
■ ■■■ generate_report.py [Self-contained HTML report dashboard generator]
■■■ runs/ [Run-specific model logs, predictions & metrics]
■■■ models/ [Run-specific saved model & pipeline joblib artifacts]
■■■ results/ [Run-specific model comparison CSV/JSON & test predictions]
■■■ reports/ [Interactive HTML Dashboard (model_comparison.html)]
```

3. TOF Component Directory Breakdown

The table below details every directory component in the final workspace architecture, summarizing contents, formats, and roles:

Directory Path (TOF)	Subdirs & Files	Primary Formats	Operational Purpose in Architecture
NeuroADHD/	8 subdirs, 4 files	.venv, data, src, tests	Original project workspace containing data lake, raw ROI files, Fisher-z feature matrix, and preprocessing logic.
data/adhd200/features/cc200/	5 files (~42.2 MB)	cc200_connectivity_features.csv	Primary ML dataset storing 153 subjects × 19,900 Fisher-z connectivity edge features + diagnosis labels.

NeuroADHD_ModelBenchmark/	7 subdirs	configs, model_envs, scripts, reports	Isolated benchmark workspace executing multi-model training, hyperparameter tuning, ensembling, and HTML reporting.
model_envs/	7 isolated envs	Python 3.11 virtualenvs, lockfiles	Model-specific isolated virtual environments ensuring strict package isolation for every model family.
scripts/	6 Python scripts	run_all_models.py, train_model.py...	Execution engine handling preflight validation, isolated model fitting, out-of-fold stacking, and HTML generation.
reports/	1 HTML report	model_comparison.html	Self-contained interactive HTML dashboard containing leaderboard, best model callouts, confusion matrices, and test results.

4. Benchmark Results Summary & Verification

Model Selection Result:

• **Best Validation Model:** Random Forest Classifier (Validation ROC-AUC = 0.7333, F1 = 0.1538).

• **Final Test Result (Peking_2 Site, 20 subjects):** Accuracy = 0.5500, Balanced Accuracy = 0.5500, Specificity = 0.8000, ROC-AUC = 0.6300.

• **Ensemble Comparison:** Soft Voting Ensemble achieved Val ROC-AUC = 0.7111; Stacking Meta-Classifier achieved Val ROC-AUC = 0.6889.

• **Data Safety Integrity:** Zero files deleted, overwritten, or modified in the original NeuroADHD repository.

• **Interactive Dashboard:** Rendered cleanly at NeuroADHD_ModelBenchmark/reports/model_comparison.html.